STRUCTURA: A Bayesian Network Framework for Information-Theoretic Scoring and Behavioral Anomaly Detection

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01 · INTRODUCTION

Under network psychometrics, constructs are modeled as interacting nodes linked by explicit dependencies. Bayesian Networks enable exact person-level inference[1,2,3], but their use in cognitive assessment remains limited.**No validated framework currently turns a single response pattern into a reliable, interpretable, and clinically meaningful score[4].** STRUCTURA would address this gap.

03 · RESULTS — THE GUTTMAN SCALE PARADOX

A Guttman paradox[6]: failing easy items while solving difficult ones.

57.7% - <5th RAW ACCURACY & NORMATIVE CENTILE	**52.9% - <1st** TESS CI 49.9–56.5 & NORMATIVE CENTILE
98.48th GSI - CI 96.97–100	**3 + 7** CARELESS + LUCKY FLAGS

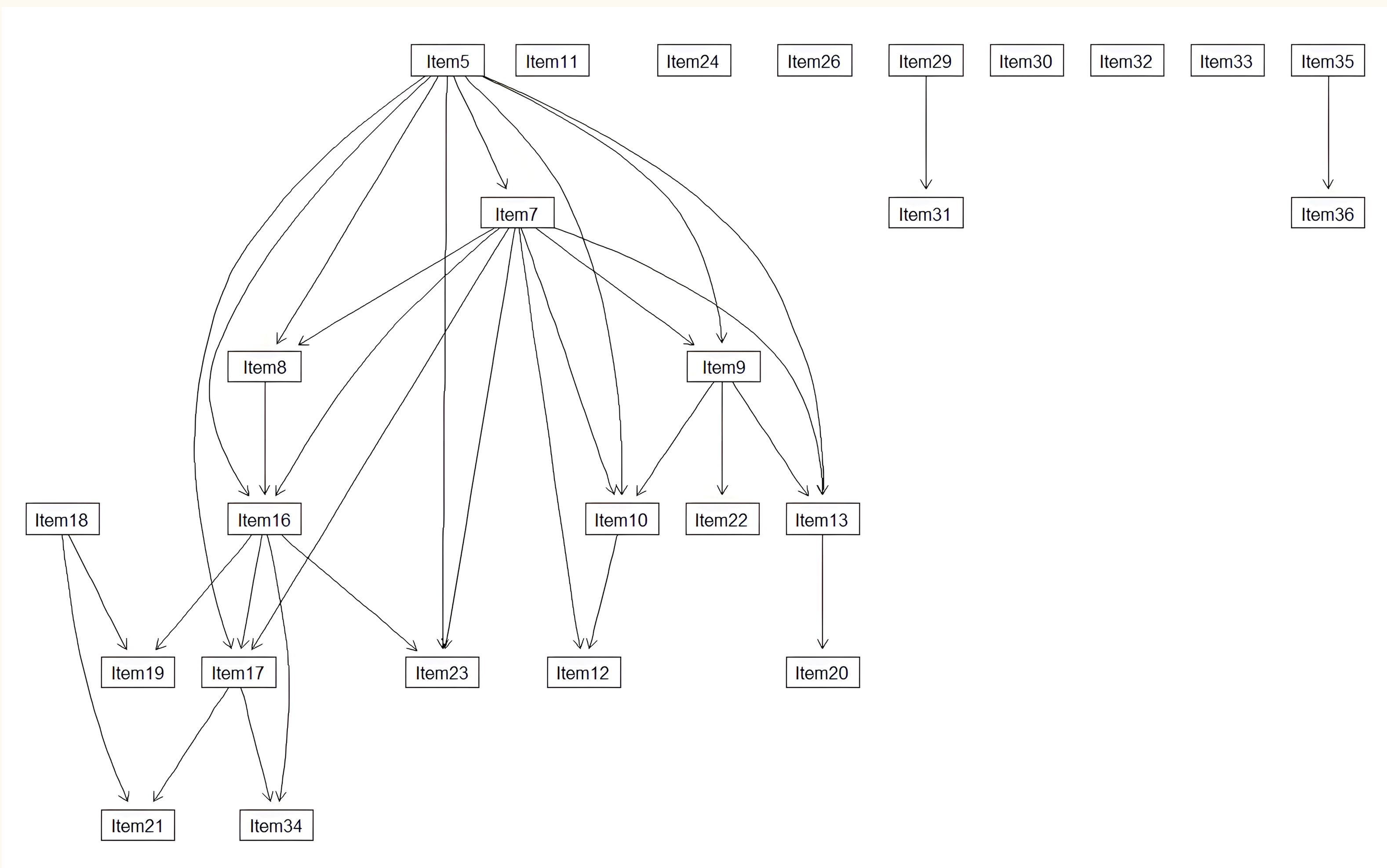

Fig. 1 — Learned normative BN of Raven's CPM, adapted from Orsoni et al. (2025)[5].

02 · METHODS

Normative sample & instrument. Normative data: 32 kindergarten children (*M* age = 4.72, *SD* = 0.30; 46.9% male) from Orsoni et al. (2025)[5]. Raven's CPM, a non-verbal measure of fluid intelligence and analogical reasoning, includes 36 dichotomous items across subscales A, Ab, and B; after excluding zero-variance items, 26 items were analysed.

2 **Compilation & local evidence.**The BN was compiled into a junction tree for exact inference, with Laplace-smoothed CPTs (α = 0.5). For each item, local evidence was defined by its Markov Blanket: $E_i = \mathrm{MB}(X_i)$. Only structurally relevant neighbours informed person-level inference.

3 **TESS.** $\sum_i P(X_i = 1 \mid E_i)$, normalised to 0–100. Equal raw accuracy can yield different TESS values.

4 **Global SI (GSI).** $\sum_i -\log_2 P(X_i = x_i \mid E_i)$, mapped onto an empirical 0–100 normative centile scale.

5 **Local SI (LSI).** $\mathrm{SI}_{\pm,i} = (x_i - E[X_i \mid E_i]) \cdot I(x_i)$: signed item-level anomaly; flagged at the 5th/95th normative percentiles.

6 **Uncertainty.** Parametric bootstrap (95% CI): each CPT perturbed via Dirichlet sampling (ISS as hyperparameter).

04 · DISCUSSION & CONCLUSIONS

These three metrics capture complementary aspects of anomaly:**TESS** adjusts the ability estimate for structural inconsistency,**GSI** quantifies how improbable the overall response pattern is, and **LSI** identifies the specific items driving the anomaly.

STRUCTURA translates BN topology into individual-level scoring.

Uncertainty is rigorously quantified via Dirichlet bootstrap CIs.

Generalizable to any assessment where a DAG is theoretically or empirically justified.

Next steps: Formally validate TESS, GSI, and LSI; integrating contextual nodes and evaluating clinical feasibility in psychiatric populations.

03B · RESULTS — LOCAL SI PROFILE (GUTTMAN CASE)

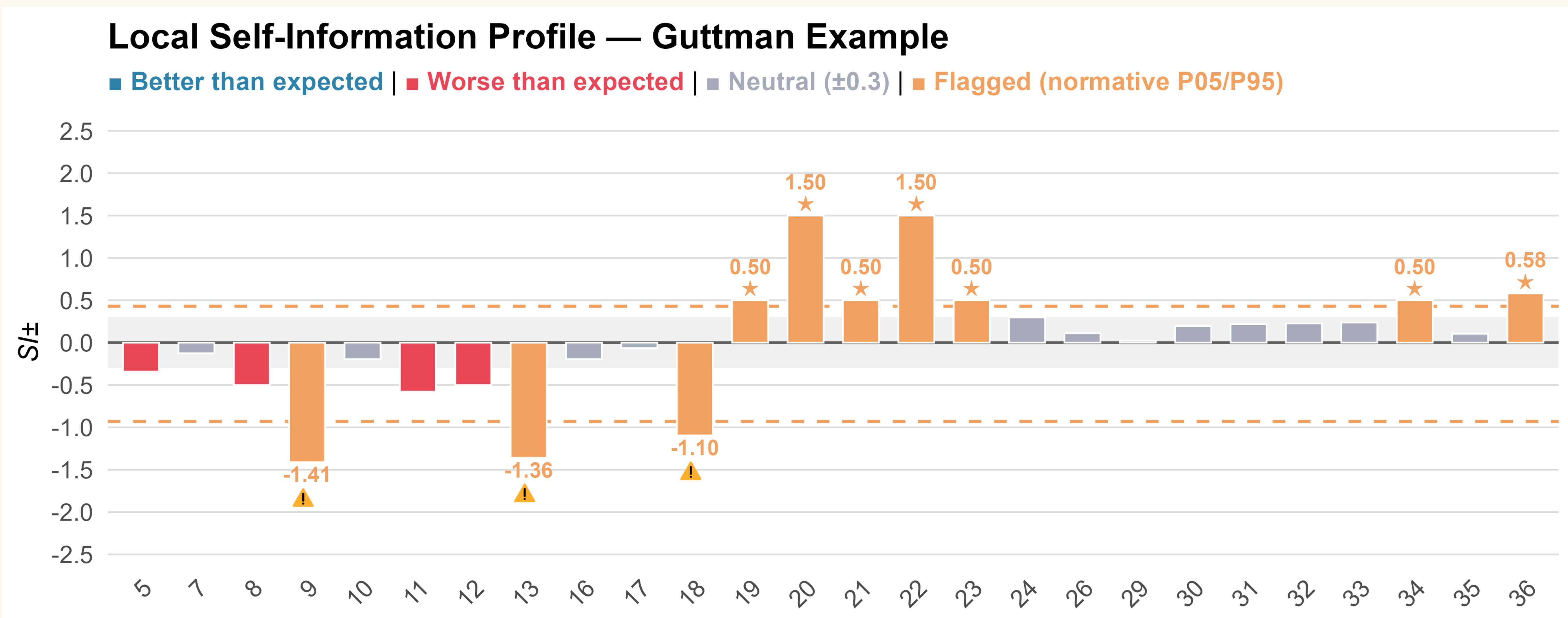

Fig. 2 — Signed SI per item; flags at normative P05 = −0.929 and P95 = 0.430.

QR CODE

Scan for full references, and poster PDF.

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